

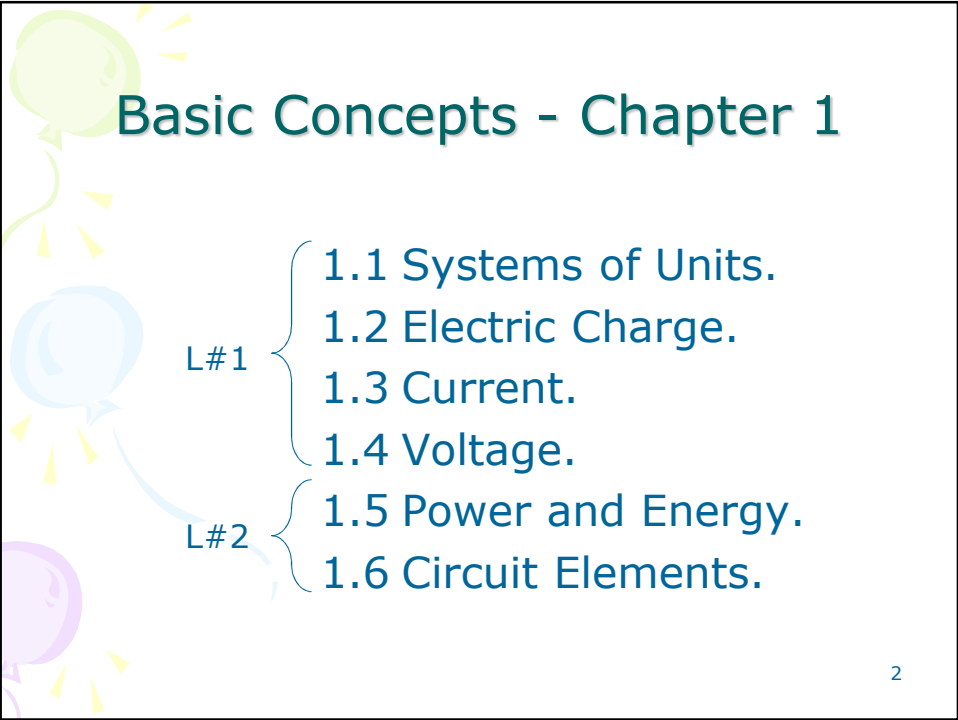
**EBN 111/122
ELECTRICITY AND
ELECTRONICS**

Chapter 1

Basic Concepts

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Basic Concepts - Chapter 1

- L#1 {
 - 1.1 Systems of Units.
 - 1.2 Electric Charge.
 - 1.3 Current.
 - 1.4 Voltage.
- L#2 {
 - 1.5 Power and Energy.
 - 1.6 Circuit Elements.

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Chapter 1: Lecture #1

1.1 System of Units (1)

Six basic units: SI Units

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd

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1.1 System of Units (2)

The derived units commonly used in electric circuit theory

Decimal multiples and submultiples of SI units

Quantity	Unit	Symbol
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
conductance	siemens	S
inductance	henry	H
capacitance	farad	F
frequency	hertz	Hz
force	newton	N
energy, work	joule	J
power	watt	W
magnetic flux	weber	Wb
magnetic flux density	tesla	T

Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

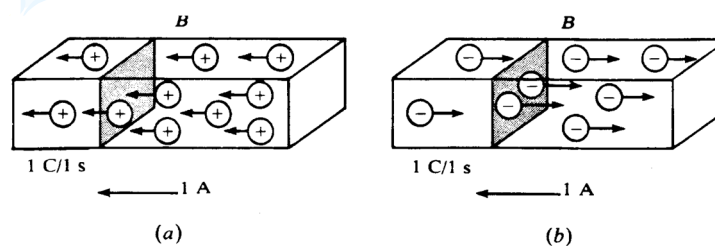
1.2 Electric Charge

- **Charge** is an electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**.
- The charge **e** on one electron is negative and equal in magnitude to $1.602 \times 10^{-19} \text{ C}$ which is called an electronic charge.
- $1 \text{ C} = 1/e = 6.24 \times 10^{18}$ electrons
- The charges that occur in nature are **integral multiples** of the electronic charge **e**.
- **Law of conservation of Charge:** Charge can neither be created nor destroyed, it can only be transferred.

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1.3 Current (1)

- The direction of current flow
 - Motion of charge creates electric current.
 - Benjamin Franklin (1706 – 1790): Convention: Current flows in the direction of **positive charge**.
 - Years later it was found that it is actually Free Electrons moving in the opposite direction.



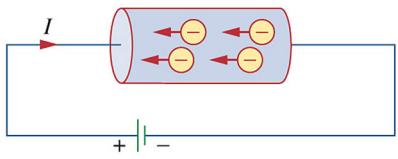
Positive ions

Negative ions

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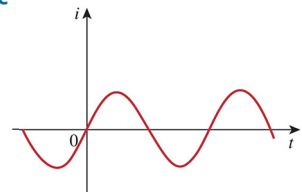
1.3 Current (2)



- Electric current I . The unit of ampere can be derived as $1 \text{ A} = 1 \text{ C/s}$.
- Amount of charge transferred between t_0 and t :
- A **direct current (dc)** is a current that remains constant with time.
- An **alternating current (ac)** is a current that varies sinusoidally with time. (reverse direction)



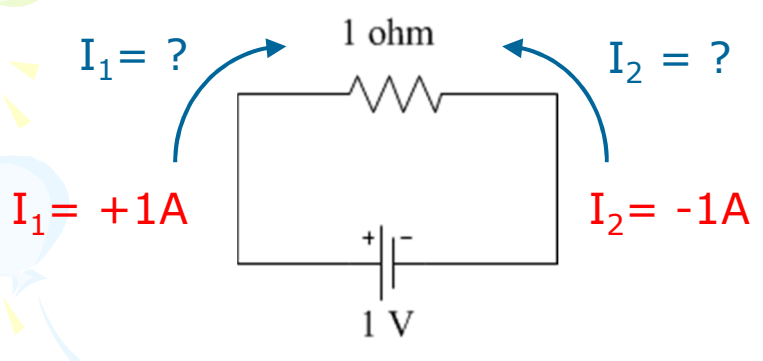
(a)



(b)

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1.3 Current (3)



$I_1 = ?$

$I_2 = ?$

$I_1 = +1\text{A}$

$I_2 = -1\text{A}$

1 ohm

1 V

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PP1.1

Calculate the amount of charge represented by four million protons.

$$\begin{aligned}
 Q &= 4 \times 10^6 \times -e \\
 &= 4 \times 10^6 \times -(-1.602 \times 10^{-19} \text{C}) \\
 &= +6.408 \times 10^{-13} \text{C} \\
 &= 0.6408 \times 10^{-12} \text{C} \\
 &= 0.6408 \text{ pC} \\
 &= 640.8 \text{ fC}
 \end{aligned}$$

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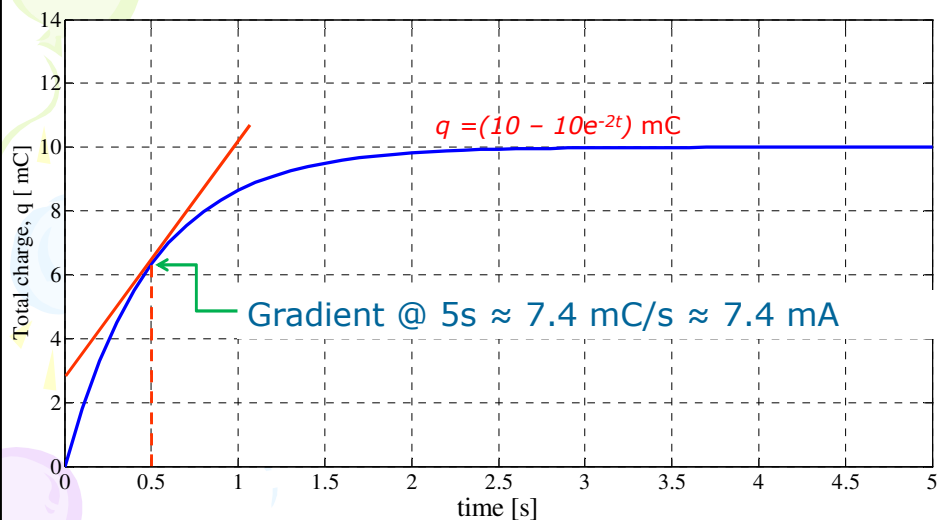
PP1-2

The total charge entering a terminal is given by $q = (10 - 10e^{-2t}) \text{ mC}$. Find the **current** at $t = 0.5 \text{ s}$.

$$i = \frac{dq}{dt}$$

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Differentiating means finding the slope or gradient



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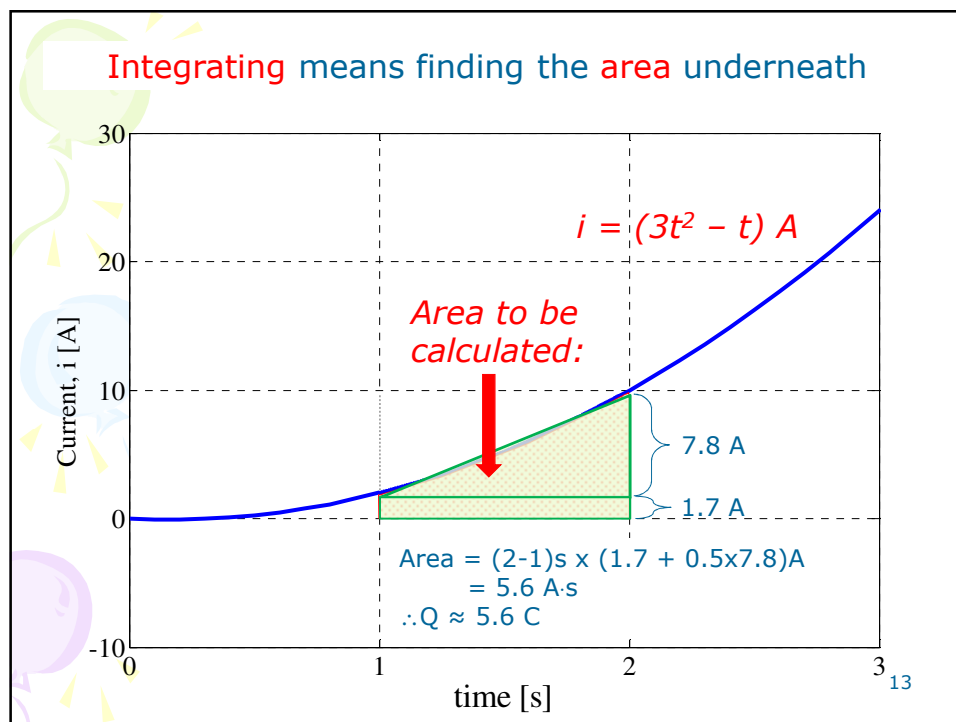
Example 1-3

Determine the total **charge** entering a terminal between $t = 1 \text{ s}$ and $t = 2 \text{ s}$ if the current passing the terminal is:

$$i = (3t^2 - t) \text{ A}$$

$$Q = \int_{t_0}^{t_1} i \, dt$$

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1.4 Voltage

- Voltage (or potential difference) is the **energy** required to move a **unit charge** through an element, measured in volts (V).
- Work is done by the electromotive force (**emf**) aka **voltage** or **potential difference**.
- Mathematically, (volt)
 - w is energy in joules (J) and q is charge in coulomb (C).
 - voltage is change in energy per charge

$$v_{ab} = dw / dq$$

$$v_{ab} = v_a - v_b$$

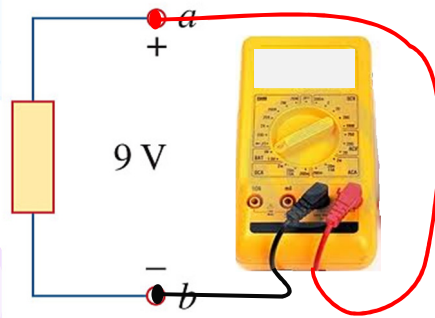
1.4 Voltage

Electric voltage, v_{ab} , is always **across the circuit element** or **between two points in a circuit**.

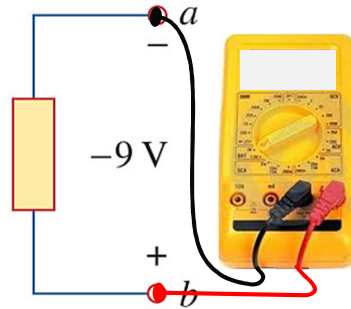
$v_{ab} > 0$ means the potential of **a** is higher than potential of **b**.

$v_{ab} < 0$ means the potential of **a** is lower than potential of **b**.

$$v_{ab} = -v_{ba}$$



(a) $v_{ab} =$



(b) $v_{ba} =$

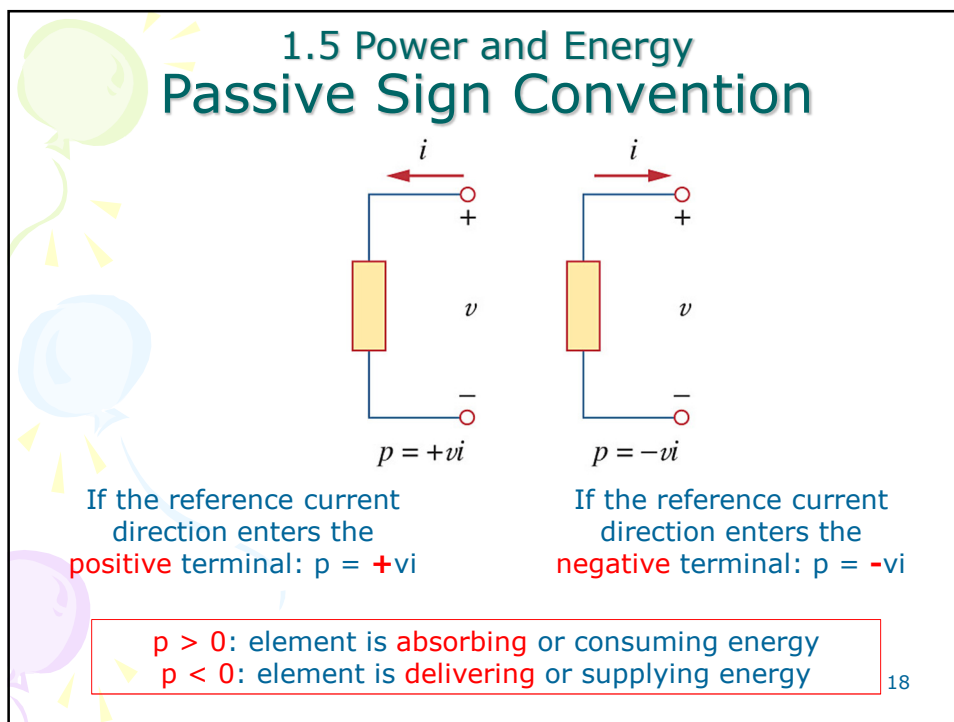
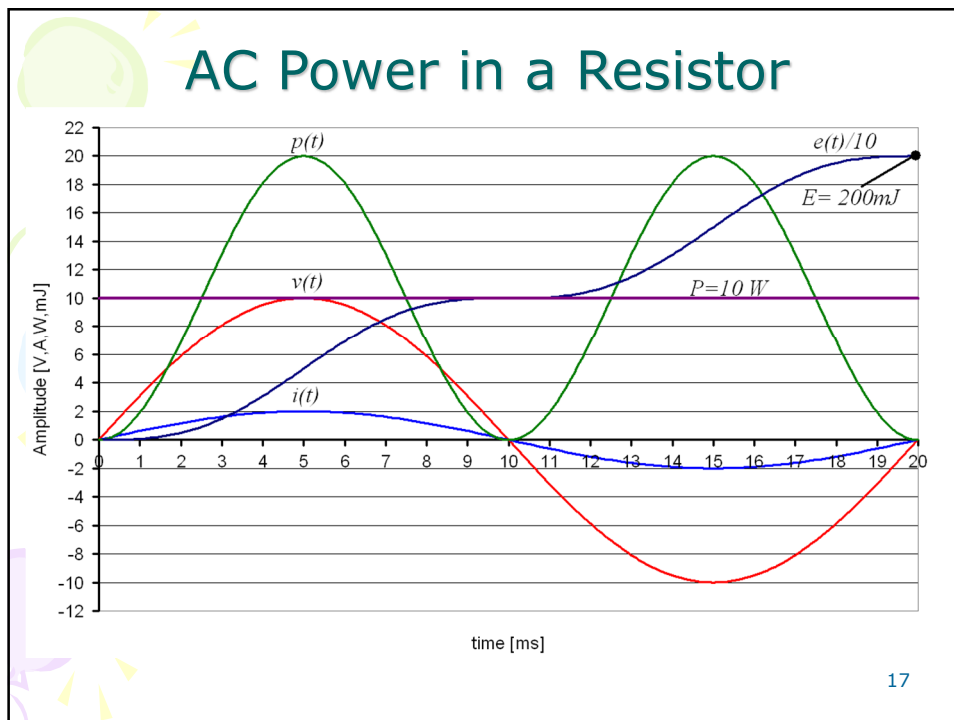
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Chapter 1: Lecture #2

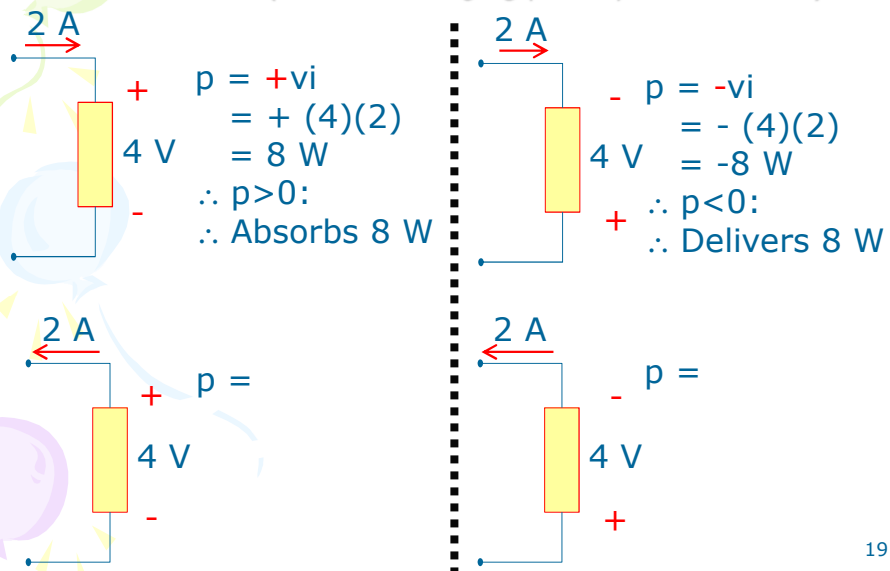
1.5 Power and Energy

- **Power** is the **rate** at which **energy** is consumed or delivered: Joules per second = Watt; $[J/s = W]$.
- Mathematical expression:
$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$
- **Instantaneous power** = the rate at which energy is consumed at a specific point in time:
- DC systems: $p(t) = P = VI = \text{constant}$
- AC systems: $p(t) = v(t)i(t) = \text{time function}$

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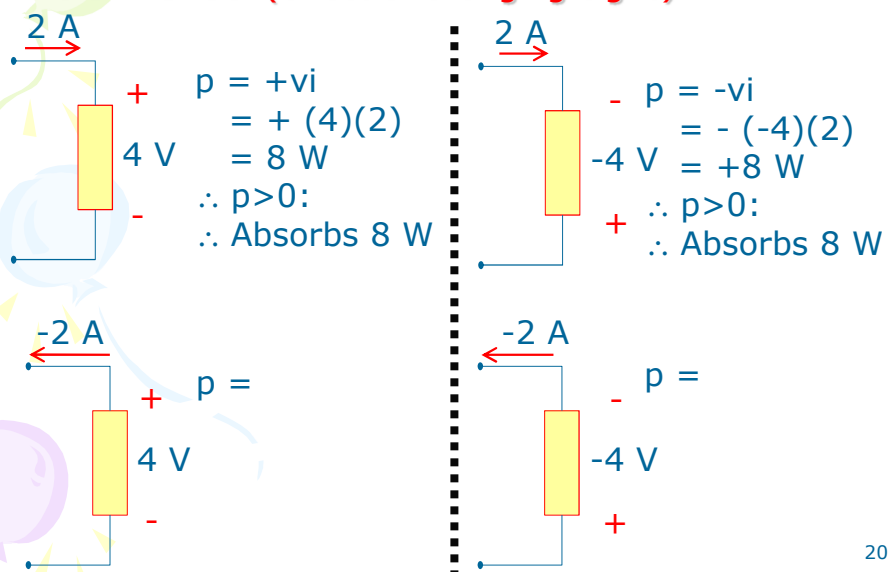


Calculate the power being absorbed or delivered (effect of changing polarity and direction):



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Calculate the power being absorbed or delivered (effect of changing signs)



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1.5 Power and Energy

- The law of **conservation of energy** implies that in an electrical circuit $p_{\text{delivered}} = p_{\text{absorbed}}$, or:

$$\sum p = 0$$

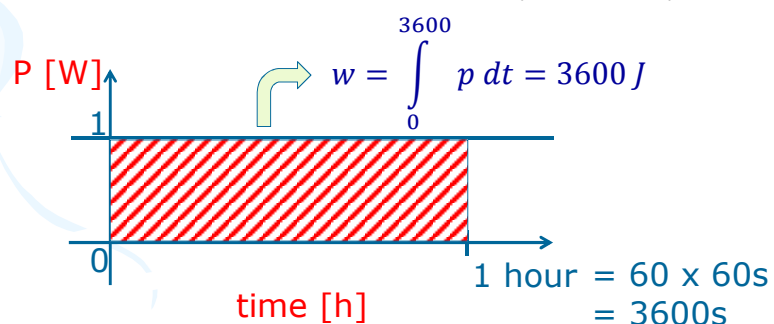
- Energy** is the capacity to do work, measured in joules (J) or alternatively in Watt-hour (Wh).
 - 1 Wh = 3600 J = 3.6 kJ
 - 1 kWh = 3.6 MJ

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1.5 Power and Energy

- Energy absorbed or delivered by an element between times t_0 and t_1 is:

$$w = \int_{t_0}^t p dt = \int_{t_0}^t v i dt$$



An element consumes energy at a rate of 1 W for 1 hour – how much energy is that in Joule?

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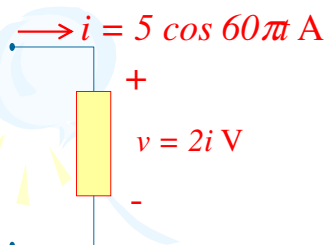
PP1-4

To move charge $q = 2\text{C}$ from point a to b requires -30 J . Find the voltage drop V_{ab} .

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PP1-5

Find the power delivered to an element at $t = 5\text{ ms}$ if the voltage is $v = 2i\text{ V}$ the current entering its positive terminal is: $i = 5 \cos 60\pi\text{ A}$.



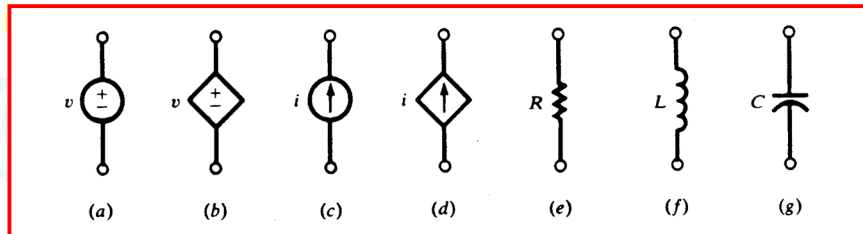
*NB: Calculator must be set to **radians!***
 $2\pi [\text{rad}] = 360^\circ$

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1.6 Circuit Elements (1)

Active Elements (Energy Sources)
Generators, batteries, op-amps

Passive Elements
Resistors, Inductors, Capacitors



Independent sources Dependent sources

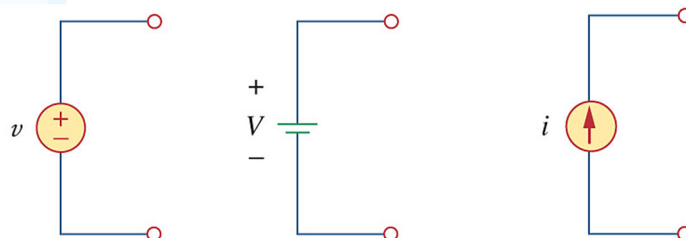
- A dependent source is an active element in which the source quantity is controlled by another voltage or current.
- They have four different types: VCVS, CCVS, VCCS, CCCS. Keep in mind the signs of dependent sources.

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Ideal Independent Sources:

An active element that provides a specified voltage or current that is completely independent of other circuit elements. **ROUND** in shape.

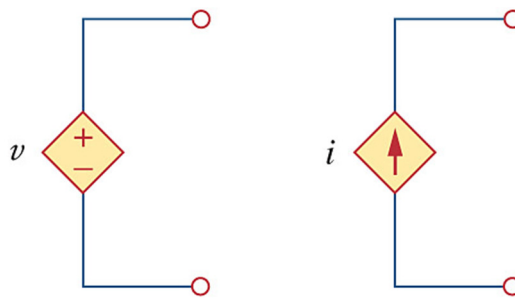
I.e. If $v = 10\text{ V}$ it stays 10 V no matter what!
If $i = 5\text{ A}$ it stays 5 A no matter what



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Ideal **Dependent** Sources:

- An active element in which the source quantity is controlled by another voltage or current. DIAMOND in shape.
- Useful in modeling elements such as transistors, operational amplifiers and integrated circuits.

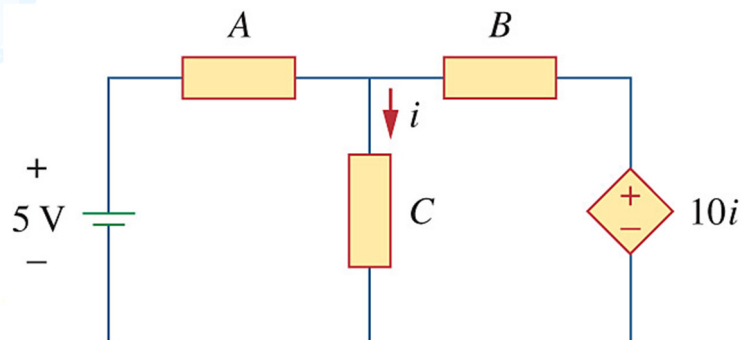


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Types of **Dependent** Sources:

Four possible types of dependent (or controlled) sources:

- VCVS = Voltage Controlled Voltage Source
- CCVS = Current Controlled Voltage Source
- VCCS = Voltage Controlled Current Source
- CCCS = Current Controlled Current Source



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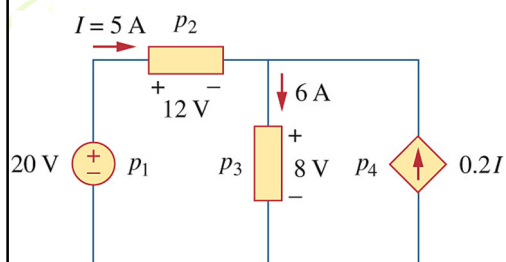
PP1-6

A stove element draws 15 A when connected to a 240 V supply. How long does it take to consume 60 kJ?

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Example 1.7

Calculate the power supplied or absorbed by each element.



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1.5 Power and Energy

Passive Sign Convention

BEWARE!

To relate power and energy to voltage and current, we recall from physics that:

Power is the time rate of expending or absorbing energy, measured in watts (W).

We write this relationship as

$$p \triangleq \frac{dw}{dt} \quad (1.5)$$

where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s). From Eqs. (1.1), (1.3), and (1.5), it follows that

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v i \quad (1.6)$$

or

$$p = v i \quad (1.7)$$

The power p in Eq. (1.7) is a time-varying quantity and is called the *instantaneous power*. Thus, the power absorbed or supplied by an element is the product of the voltage across the element and the current through it. If the power has a $+$ sign, power is being delivered to or absorbed by the element. If, on the other hand, the power has a $-$ sign, power is being supplied by the element. But how do we know when the power has a negative or a positive sign?

Current direction and voltage polarity play a major role in determining the sign of power. It is therefore important that we pay attention to the relationship between current i and voltage v in Fig. 1.8(a). The voltage polarity and current direction must conform with those shown in Fig. 1.8(a) in order for the power to have a positive sign.

This is known as the *passive sign convention*. By the passive sign convention, current enters through the positive polarity of the voltage. In this case, $p = +vi$ or $vi > 0$ implies that the element is absorbing power. However, if $p = -vi$ or $vi < 0$, as in Fig. 1.8(b), the element is releasing or supplying power.

Passive sign convention is satisfied when the current enters through the positive terminal of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$.

Unless otherwise stated, we will follow the passive sign convention throughout this text. For example, the element in both circuits of Fig. 1.9 has an absorbing power of $+12$ W because a positive current enters the positive terminal in both cases. In Fig. 1.10, however, the element is supplying power of $+12$ W because a positive current enters the negative terminal. Of course, an absorbing power of -12 W is equivalent to a supplying power of $+12$ W. In general,

$+ \text{Power absorbed} = - \text{Power supplied}$

The extracted text on p 11 is very misleading as it is not stated accurately.

Please also avoid Fig. 1.8, 1.9 and 1.10 as they will create confusion regarding the passive sign convention and as to when power is being supplied or absorbed.

Please refer to the notes and examples provided in class. Specifically slides 18, 19 and 20 of Chapter 1 (Lecture #2)

This is known as the *passive sign convention*. By the passive sign convention, current enters through the positive polarity of the voltage. In this case, $p = +vi$ or $vi > 0$ implies that the element is absorbing power. However, if $p = -vi$ or $vi < 0$, as in Fig. 1.8(b), the element is releasing or supplying power.

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